

How to compute $x(t)$ and $y(t)$ in practice?

Last time we've focused on x_{free} . We've seen increased complexity in the computation of the term. We've reached a point in which we were asking how to find T such that

$$AT = T\Lambda$$

We've seen how we should rewrite T highlighting the columns, getting the following condition:

$$Au_i = \lambda_i u_i, i = 1, \dots, n$$

This is the key necessary property for a T matrix that can decouple the system. u_i must be eigenvectors of matrix A , and λ_i are the associated eigenvalues.

The opposite is also true: if I can find n independent eigenvectors, then I can form a matrix T which can decouple the system.

Key problem is this: find eigenvectors and eigenvalues of a matrix A .

The key equation to find eigenvectors and eigenvalues is, in simple words, about finding any vector u and number λ satisfying the following equation:

$$\begin{aligned} Au &= \lambda u \\ \rightarrow Au - \lambda u &= 0 \rightarrow (A - \lambda I)u = 0 \end{aligned}$$

The last one is called the characteristic equation of system.

$(A - \lambda I)$ is a $n \times n$ matrix, and u is a vector of elements, of dimension $n \times 1$.

We are looking for non trivial solutions, such that $u \neq 0$.

The key condition for non-trivial solutions ($u \neq 0$) is that the $\det((A - \lambda I)) = 0$.

By computing $\det(\cdot)$ we get a polynomial of degree n , written in this way:

$$1\lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_1\lambda + a_0 = 0$$

This equation takes the name of a characteristic equation. It has n solutions, and they can either be real numbers and/or complex numbers. Also, these solutions can be distinct, or (some or all) coincident.

≡ Example

$$d(\lambda) = (\lambda - 1)(\lambda + 2)(\lambda^2 + 1)(\lambda + 4)^3$$

Here we have 7 solutions, as for $\lambda_1 = 1$, $\lambda_2 = -2$, $\lambda_{3,4} = \pm\sqrt{-1}$ and

$$\lambda_{5,6,7} = -4.$$

We could plot this on the "ascissa" to be set to Real numbers, and the "ordinata" to Imaginary numbers.

Simplest Case:

$d(\lambda)$ has all solutions that are real values and distinct.

Thus, I will find n solutions $(\lambda_1, \dots, \lambda_n)$ of $\det(A - \lambda I) = 0$. Sometimes, $|A - \lambda I| = 0$ is used to indicate the $\det(\cdot)$.

We have n real and distinct eigenvalues.

In this case it's possible to prove that I will find n linearly independent eigenvectors. By doing this, I will be able to form the columns of T matrix.

Next Step:

Then, given a generic λ_i , we must find the associated eigenvector by solving:

$$(A - \lambda_i I)u = 0 \quad \text{with } \lambda_i \text{ number, } u \text{ to be found}$$

So, to recap:

1. Find the eigenvalues by solving $\det(A - \lambda I) = 0$. In the simple case, we find $\lambda_1, \dots, \lambda_n$ real and distinct eigenvalues.
2. For every eigenvalue λ_i , $i = 1, \dots, n$, find the associated eigenvector $u_1, u_2, \dots, u_n$ by solving $(A - \lambda_i)u = 0$, with u to be found.
3. T will be equivalent to $[u_1, u_2, \dots, u_n]$.

Back to the solution formula, we introduce the concept of Natural Modes of a system.

Last lesson we've seen that $x(t) = Te^{\Lambda(t-t_0)}T^{-1}x(t_0)$. Now, write $T = [u_1, \dots, u_n]$ and also highlight the rows of T^{-1} , obtaining:

$$T^{-1} = \begin{bmatrix} v_1^T \\ v_2^T \\ \dots \\ v_n^T \end{bmatrix}$$

(We're using the T notation as typically, in this course, when we write a vector it's a column vector, but now we're highlighting rows).

By putting these expressions in the formula for $x(t)$ we get this:

$$x(t) = \dots = e^{\lambda_1(t-t_0)}u_1v_1^T x(t_0) + e^{\lambda_2(t-t_0)}u_2v_2^T x(t_0) + \dots + e^{\lambda_n(t-t_0)}u_nv_n^T x(t_0) =$$

$$= \sum_{i=1}^n e^{\lambda_i(t-t_0)} u_i v_i^T x(t_0)$$

The free evolution of a system can be decomposed as the sum of n independent terms.

If we focus on the "extended formula", we see that $e^{\lambda_1(t-t_0)}$ is ?, u_1 is a column vector, v_1 is a row vector multiplied by a column vector, from which we get a number, let's define it as c_i .

So, we can write (having $t_0 = 0$):

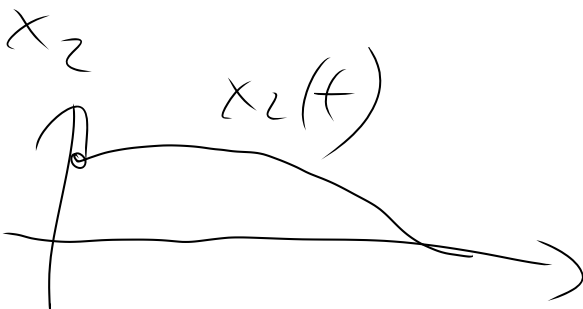
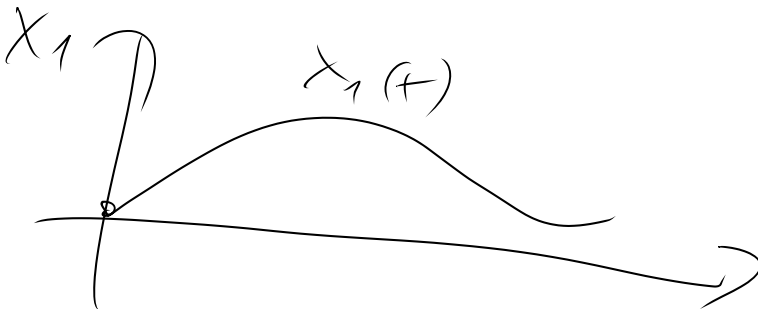
$$x_{free}(t) = c_1 \cdot e^{\lambda_1 t} u_1 + c_2 e^{\lambda_2 t} u_2 + \dots + c_n e^{\lambda_n t} u_n$$

This tells us how the system will evolve in time.

What are these c coefficients? We know their formula.

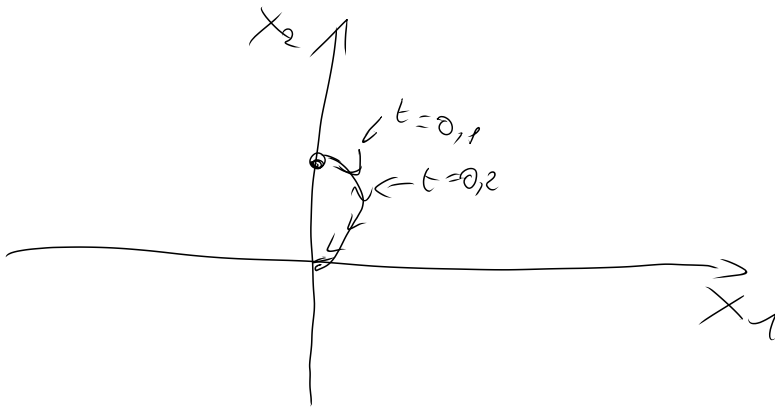
Write now $x_{free}(0)$. We have that $x_{free}(0) = x(0)$, as the initial solution is the initial state. So, $x(0) = c_1 u_1 + c_2 u_2 + \dots + c_n u_n$.

Let's see a practical example. Suppose we have a RLC circuit, with $x = \begin{bmatrix} i_L \\ v_c \end{bmatrix}$.



The formula gives us exactly this.

Now, if we put together the plots, having x_1 and x_2 as axis:

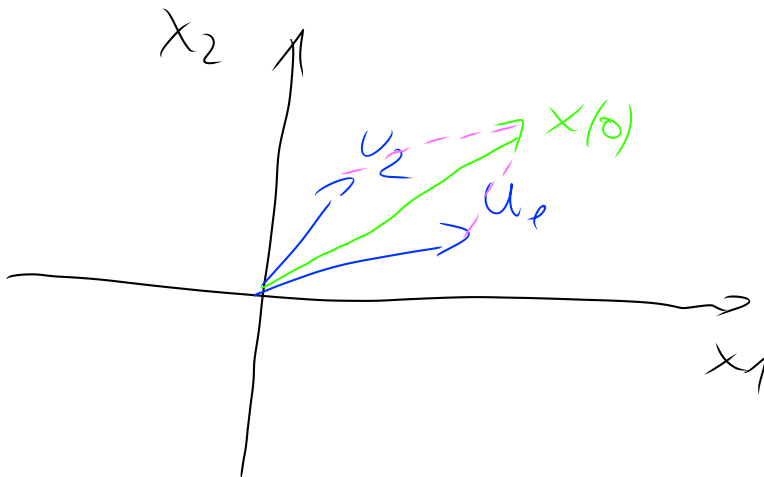


The arrows tell us where the state moves. We're drawing the trajectory of the state in time.

Let's go back now to

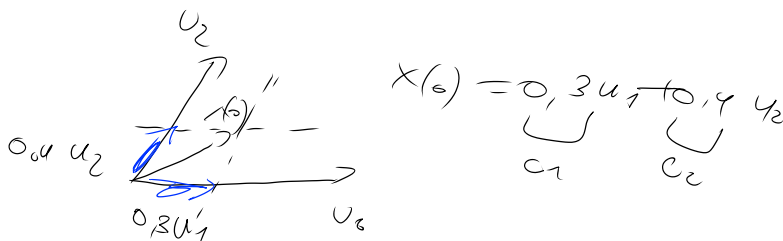
$$x(0) = c_1 u_1 + c_2 u_2 + \dots + c_n u_n$$

We're writing $x(0)$



with new coordinates, based on u_1 and u_2 .

c coefficients are the coordinates of the initial state $x(0)$ in the new reference system given by $u_1, u_2, \dots, u_n$.



Let's say we pick $x(0)$ aligned with u_1 . In this case the system is decoupled, moving along just one eigenvector.

We have that $e^{\lambda_i t} u_i$ are called the natural modes of the system. In particular, $e^{\lambda_i t}$ is the temporal law of natural mode.

In a simple case on n real and distinct eigenvalues, I have n distinct exponential natural modes.

Let's take a look now at exponential natural modes, focusing on temporal law.

I can have three cases:

Case 1 is given by $\lambda_i < 0$, so a real and negative eigenvalue. Then, the plot of $e^{\lambda_i t}$ is decreasing and goes to zero. We can call this stable, converging case.

Case 2 is given by $\lambda_i = 0$. Then, the value is constant.

Case 3 is given by $\lambda_i > 0$, which leads to a diversion of the value.

What happens when λ_i are distinct, but some of them (or all) are not real numbers?

For example, we have $\lambda_1 = 1$, $\lambda_2 = -3$, $\lambda_3 = 1 + 3i$ and $\lambda_4 = 1 - 3i$.

Then, we have:

$$x_{free}(t) = c_1 e^{\lambda_1 t} u_1 + c_2 e^{\lambda_2 t} u_2 + \text{---} + \text{---}$$

the last two values are pseudoperiodic modes.

We have that $\lambda_{3,4}$ corresponds, indeed, to pseudoperiodic modes.

By plotting time on x -axis, and the temporal law of pseudoperiodic mode on the y -axis, we can see that there are oscillations. The temporal mode oscillates between two envelope ? that are exponentials of the real part of the value $e^{\text{Re}(\lambda_i)t}$. When we have that $\text{Re}(\lambda_i) > 0$ we have a diverging pseudoperiodic mode.

Case 2 would be given by having $\text{Re}(\lambda_i) = 0$, for example $\lambda_{3,4} = 0 \pm i$.

The temporal law is now a periodical signal, not pseudo-periodical.

Finally, case 3 is given by $\text{Re}(\lambda_i) < 0$, with oscillations that decrease in time and go to zero. We call this a converging pseudo-periodic mode.

The RLC circuit we've studied, depending on parameters, may have oscillations or go to zero, for example.

What is the last case now? For now we've seen that when λ_i is a real number, we have an exponential natural mode. When λ_i is complex we have a pseudo-periodic mode.

We also have another mode, which can arise in a particular case, when some (or all) eigenvalues are coincident.

For example:

Suppose we get to compute $\det(A - \lambda I) = 0 \rightarrow (\lambda - 1)(\lambda + 2)^2 = 0$. Now we have

$\lambda_1 = 1, \lambda_2 = -2, \lambda_3 = -2$. λ_1 will give me one eigenvector u_1 , but for the other two I have to solve $(A + 2I)u = 0$. We can have two cases:

1. I find two distinct eigenvectors, so I can form $T = [u_1, u_2, u_3]$, and
$$x_{free} = c_1 e^{\lambda_1 t} u_1 + c_2 e^{\lambda_2 t} u_2 + c_3 e^{\lambda_3 t} u_3.$$
2. I find only one eigenvector for -2 . In T we'll have u_1, u_2 , and then another object known as generalized eigenvector, which will give birth to the third case of natural mode, the polynomial natural mode.